

Correlation-driven topological phases in magic-angle twisted bilayer graphene

<https://doi.org/10.1038/s41586-020-03159-7>

Received: 21 August 2020

Accepted: 13 November 2020

Published online: 18 January 2021

 Check for updates

Youngjoon Choi^{1,2,3,7}, Hyunjin Kim^{1,2,3,7}, Yang Peng⁴, Alex Thomson^{2,3,5}, Cyprian Lewandowski^{2,3,5}, Robert Polski^{1,2}, Yiran Zhang^{1,2,3}, Harpreet Singh Arora^{1,2}, Kenji Watanabe⁶, Takashi Taniguchi⁶, Jason Alicea^{2,3,5} & Stevan Nadj-Perge^{1,2,✉}

Magic-angle twisted bilayer graphene (MATBG) exhibits a range of correlated phenomena that originate from strong electron–electron interactions. These interactions make the Fermi surface highly susceptible to reconstruction when ± 1 , ± 2 and ± 3 electrons occupy each moiré unit cell, and lead to the formation of various correlated phases^{1–4}. Although some phases have been shown to have a non-zero Chern number^{5,6}, the local microscopic properties and topological character of many other phases have not yet been determined. Here we introduce a set of techniques that use scanning tunnelling microscopy to map the topological phases that emerge in MATBG in a finite magnetic field. By following the evolution of the local density of states at the Fermi level with electrostatic doping and magnetic field, we create a local Landau fan diagram that enables us to assign Chern numbers directly to all observed phases. We uncover the existence of six topological phases that arise from integer fillings in finite fields and that originate from a cascade of symmetry-breaking transitions driven by correlations^{7,8}. These topological phases can form only for a small range of twist angles around the magic angle, which further differentiates them from the Landau levels observed near charge neutrality. Moreover, we observe that even the charge-neutrality Landau spectrum taken at low fields is considerably modified by interactions, exhibits prominent electron–hole asymmetry, and features an unexpectedly large splitting between zero Landau levels (about 3 to 5 millielectronvolts). Our results show how strong electronic interactions affect the MATBG band structure and lead to correlation-enabled topological phases.

MATBG features a triangular moiré lattice with a period of $L_m = a/[2\sin(\theta/2)] \approx 13$ nm, set by the twist angle $\theta \approx 1.1^\circ$ and the graphene lattice constant $a = 0.246$ nm (refs. ^{9,10}). This additional spatial periodicity gives rise to an effective mini-Brillouin zone, which, owing to its small size, is expected to give rise to topological Chern insulating phases at realistic magnetic fields^{11–14} characteristic of a Hofstadter spectrum^{15,16}. Yet, the impact of strong correlations on Hofstadter physics and the resulting topological phases in MATBG remain underexplored. We study MATBG placed on a tungsten diselenide (WSe₂) monolayer, a thick hexagonal boron nitride (hBN) dielectric layer and a graphite gate; see Fig. 1a, Extended Data Fig. 1 and Methods section ‘Device fabrication’. We use monolayer WSe₂ as the immediate MATBG substrate because previous work¹⁷ suggested that WSe₂ improves the sample quality and does not change the magic-angle condition. In scanning tunnelling microscopy (STM), the twist angle can be determined by measuring the distance between neighbouring AA moiré sites in the topographic data (Fig. 1b)^{18–20}. The measured quantity, the differential tunnelling conductance between the tip and the sample (dI/dV) corresponding to the local density of states (LDOS), monitored as a function of the tip–sample

bias (V_{bias}) and of the gate voltage (V_{gate}) that tunes electrostatic doping, shows the evolution of the two LDOS peaks originating from the Van Hove singularities (VHSs) of the flat bands (Fig. 1c). At $V_{\text{gate}} > +5$ V, both peaks are below the Fermi energy (E_F ; $V_{\text{bias}} = 0$ mV), indicating completely filled flat bands. As V_{gate} is reduced, the first VHS corresponding to the conduction flat band crosses E_F several times, resetting its position around gate voltages corresponding to occupations of $\nu = 3, 2, 1$ electrons per moiré unit cell, until the $\nu = 0$ charge neutrality point (CNP) is reached—where the splitting between VHSs is maximized^{19–22} (see Methods section ‘Filling-factor assignment’ for the assignment of filling factor ν to V_{gate}). A similar cascade has been previously observed in MATBG placed directly on hBN⁸; our observations thus demonstrate that the added WSe₂ does not considerably alter the spectrum or the cascade mechanics (see also Methods section ‘Effects of WSe₂’).

Landau levels in MATBG

In a perpendicular magnetic field, the overall spectrum changes as Landau levels (LLs) develop in the flat bands around the CNP, and the

¹T. J. Watson Laboratory of Applied Physics, California Institute of Technology, Pasadena, CA, USA. ²Institute for Quantum Information and Matter, California Institute of Technology, Pasadena, CA, USA. ³Department of Physics, California Institute of Technology, Pasadena, CA, USA. ⁴Department of Physics and Astronomy, California State University, Northridge, CA, USA. ⁵Walter Burke Institute for Theoretical Physics, California Institute of Technology, Pasadena, CA, USA. ⁶National Institute for Materials Science, Tsukuba, Japan. ⁷These authors contributed equally: Youngjoon Choi, Hyunjin Kim. ✉e-mail: s.nadj-perge@caltech.edu

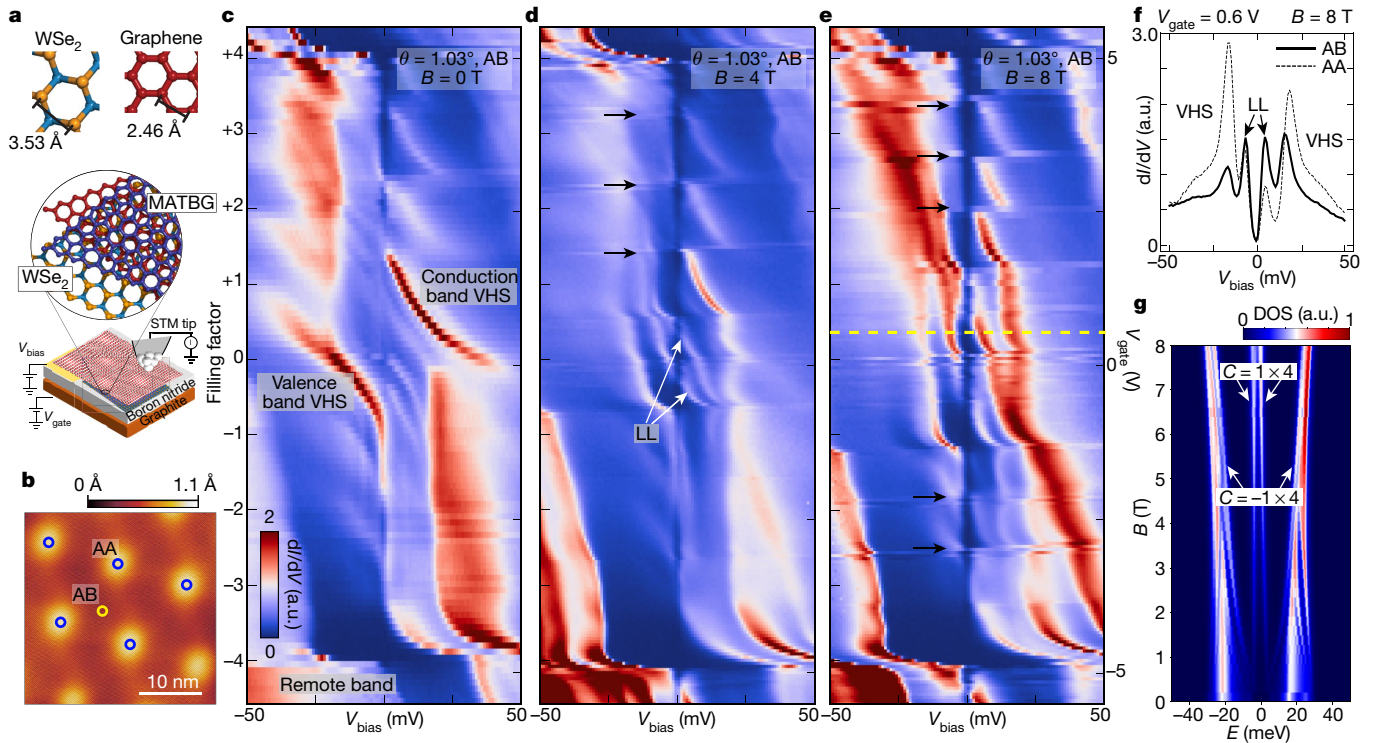


Fig. 1 | Spectroscopy of MATBG with a magnetic field at 2 K. **a**, Experimental setup. MATBG is placed on a monolayer WSe₂ and supported by hBN. A graphite gate resides underneath. The inset shows details of the structure of the WSe₂ and graphene crystal. **b**, Typical topography showing a moiré pattern at the magic angle ($V_{\text{bias}} = -400$ mV, $I = 20$ pA). **c–e**, Point spectra on an AB site at $\theta = 1.03^\circ$ as a function of V_{gate} for magnetic fields of $B = 0$ T (**c**), 4 T (**d**) and 8 T (**e**) applied perpendicular to the sample. Panel **c** shows the evolution of the zero-field density-of-states peaks originating from the flat-band VHS. As each of the peaks crosses the Fermi energy, a cascade of transitions appears, manifested here as splitting of the VHS into multiple branches close to integer filling factors ν . Features observed near the CNP are discussed further in Methods section ‘LLs and zero-field features near the CNP’. Finite-field measurements in **d** and **e** reveal LLs forming around charge neutrality ($\nu = 0$),

along with newly formed field-induced gaps (black arrows), visible as a suppression of the tunnelling conductance dI/dV . **f**, Conductance linecuts at $V_{\text{gate}} = 0.6$ V and $B = 8$ T on AA and AB sites. LLs are more visible on the AB site owing to the reduced VHS weight. Because the large intensity of the VHS peaks on the AA sites can obscure some of the fine features related to high fields, we focus on the AB sites throughout the paper (see Extended Data Fig. 2 for more spectroscopy data on the AA site). **g**, Energy spectrum calculated from the ten-band model using parameters chosen such that the relative peak positions match the experimental data in **f**. Most LLs merge into the electron and hole VHSs that each carry Chern number $C = -1$, whereas two isolated LLs at charge neutrality remain at around zero energy and both carry $C = +1$. Owing to spin–valley degeneracy, the total Chern number for zLLs and the VHS is $C = +4$ and $C = -4$. Here a.u. denotes arbitrary units.

onsets of the cascade transitions shift accordingly (see black arrows in Fig. 1d, e). These onsets are accompanied by a low LDOS at E_F and nearly horizontal resonance peaks, indicating the presence of gapped states²³. Focusing first on identifying LLs near the CNP, a linecut at $B = 8$ T shows four well resolved peaks (Fig. 1f). The phenomenological ten-band model²⁴ with parameters chosen to semiquantitatively match the data (Fig. 1g) suggests that the inner peaks are zero LLs (zLLs) originating from MATBG Dirac points, whereas the outer peaks (that include the VHS) form from the LLs that descend from less-dispersive parts of the band structure and cannot be individually resolved. Importantly, both the zLLs and the broad feature that includes the VHS are expected to be fourfold spin–valley degenerate and carry non-zero Chern number ($C = +1$ and -1 , respectively, per spin and valley flavor; see also Methods section ‘Chern number assignment for MATBG in a magnetic field’). For simplicity, we will refer to the $C = -1$ broad feature (that consists of non-resolved LLs and includes the VHS) as ‘VHS’. In this notation, the total Chern number of the zLL (VHS) is $C = +4$ ($C = -4$). The fourfold degeneracy of zLLs is further corroborated by the observation of splitting into four distinct branches when each of the zLLs crosses the Fermi energy (Extended Data Fig. 4). The Chern number assignment takes into account splitting of Dirac cones in energy, which is experimentally observed by the presence of two inner peaks instead of one, and accounts for the reduction of eightfold to fourfold degeneracy observed in previous transport MATBG experiments (see Methods

sections ‘Chern number assignment for MATBG in a magnetic field’ and ‘Quantum Hall ferromagnetism of 0^+ and 0^- LLs’). With this interpretation in mind, and as discussed in the remainder of the paper, we attribute the shifting of the zero-field cascades (black arrows in Fig. 1d, e; see also Methods section ‘Magnetic-field dependence of the cascade of phase transitions’ and Extended Data Fig. 3) and the accompanying gaps to the formation of Chern insulating phases enabled by correlations.

LDOS Landau fan diagram

The spectrum of MATBG in magnetic fields has so far been studied using transport measurements that provide information about the electronic structure close to E_F (refs.^{1,3,4,25}). We relate these experiments to our STM findings by using a novel approach that enables measuring a full Landau fan diagram via the LDOS. This LDOS Landau fan is acquired by taking the tunnelling conductance without feedback and keeping $V_{\text{bias}} = 0$ mV, such that the STM probes MATBG at E_F as the electron density (V_{gate})²⁰ and the magnetic field are varied. The resulting signal, which reflects LDOS at E_F , is thus suppressed in certain regions of carrier densities in which gaps develop (Fig. 2a, b).

Our LDOS Landau fan (Fig. 2c), taken at one specific AB point, reproduces many of the features established in previous MATBG magneto-transport measurements^{1,3,4,17}, although it records a fundamentally different quantity—LDOS instead of transport resistance.

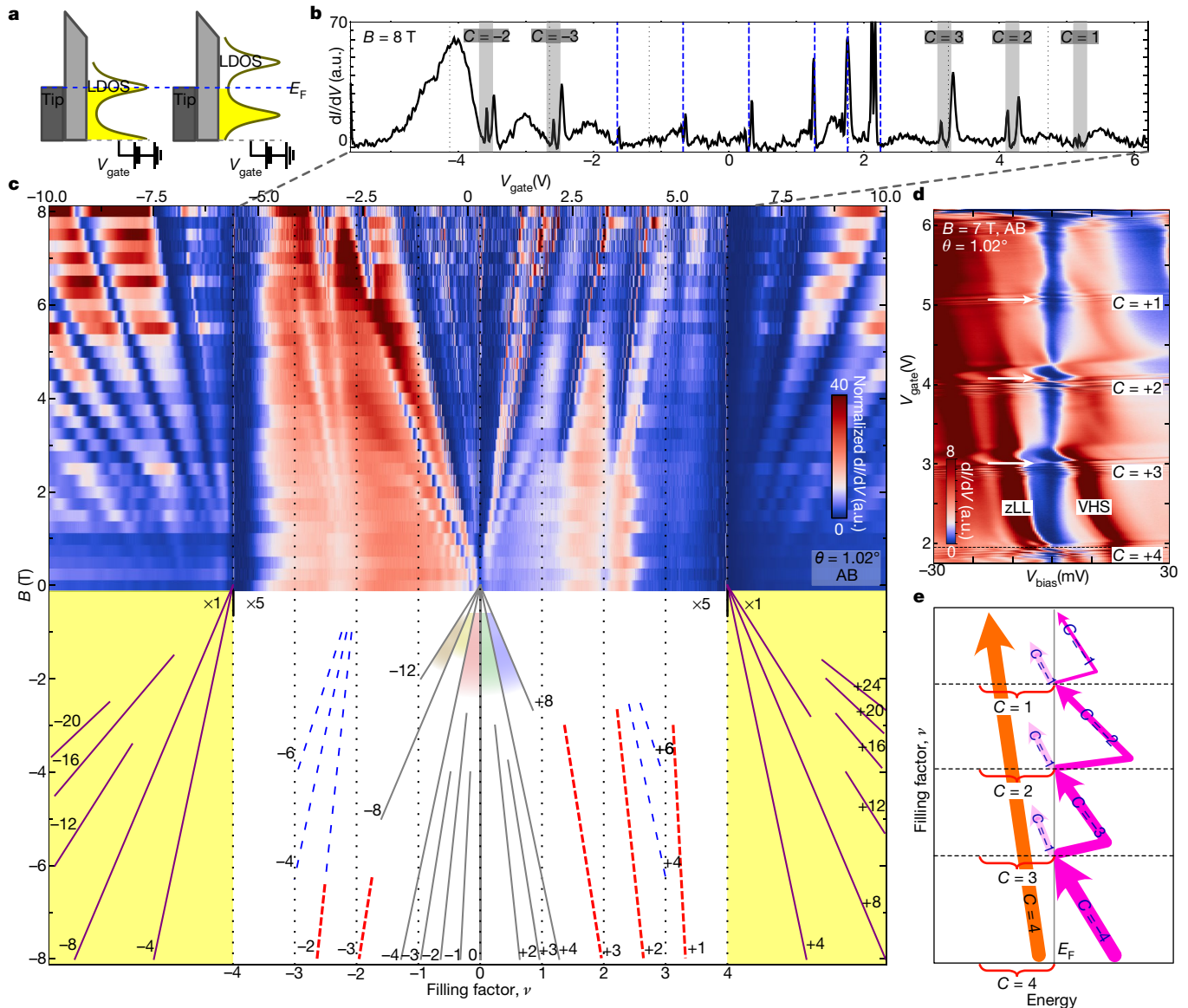


Fig. 2 | LDOS Landau fan and gaps induced by Chern insulating phases. **a**, Principle of acquiring the LDOS Landau fan. The conductance, $dI/dV \propto \text{LDOS}$, is measured while sweeping V_{gate} to change the carrier density at fixed $V_{\text{bias}} = 0 \text{ mV}$. **b**, Example linecut taken at $B = 8 \text{ T}$ and $-6 \text{ V} < V_{\text{gate}} < 6 \text{ V}$. Insulating phases appear as LDOS dips. Chern insulating phases and LLs are indicated by grey regions and blue vertical lines, respectively. The position of these lines is obtained from the slopes in **c**. **c**, LDOS Landau fan diagram at an AB site for $\theta = 1.02^\circ$. LDOS data, taken by sweeping V_{gate} , are normalized by an average LDOS value for each magnetic field (separately for flat and remote bands). Black solid (blue dashed) lines indicate gaps between LLs, originating from the CNP (half-filling). Purple lines on the yellow background show the LL gaps in remote (dispersive) bands. Magnetic-field-activated correlated Chern insulator gaps are marked by red dashed lines. The signal in the flat-band region

is multiplied by 5 to enhance visibility in relation to the remote bands. The uncertainty in V_{gate} of the $\nu = \pm 4$ position is approximately 0.1 V ; see Methods section 'Filling-factor assignment'. The colour coding of regions near the CNP separated by LL gaps marks different LLs in Fig. 4. **d**, Point spectra on the same AB point as in **c** as a function of V_{gate} taken at $B = 7 \text{ T}$, highlighting the crossing of the electron-side VHS in a magnetic field. Chern insulator gaps corresponding to $C = 3$, $C = 2$ and $C = 1$ are indicated by white arrows. The gaps are accompanied by resonances (horizontal features) originating from quantum dots formed within the insulating bulk (see Methods section 'Quantum dot formation'). Similar features are also observed on the hole side (see Extended Data Fig. 5). **e**, Schematic of the cascade in a magnetic field. Each time the VHS crosses the Fermi energy, a Chern gap appears and the corresponding band Chern number changes.

This approach also allows us to verify whether the twist-angle values extracted from the Landau fan match those seen in topography (agreement is within 0.01°). Importantly, in addition to LLs, we observe strong LDOS suppressions that abruptly appear in finite fields ($B > 3 \text{ T}$ for $\nu > 0$ and $B > 6 \text{ T}$ for $\nu < 0$; red dashed lines in Fig. 2c), indicating the formation of insulating phases emanating from $\nu = \pm 1, \pm 2, \pm 3$ (refs. ^{26–28}). The corresponding Chern numbers, $C = \pm 3, \pm 2, \pm 1$, respectively, can be directly assigned from the observed slopes of the gap positions as a function of ν and the diophantine equation²⁹, $\nu(B) = (CA_m B / \phi_0) + \nu(B = 0)$

(see Extended Data Fig. 6 for data showing the $C = -1$ state). Here, $A_m = \sqrt{3}L_m^2/2$ is the moiré unit cell area, ϕ_0 is the flux quantum and $\nu(B = 0)$ denotes the filling ($\pm 1, \pm 2, \pm 3$) from which the phases emanate.

Correlated Chern insulating phases

To better understand how Chern phases develop, we perform spectroscopic measurements for a fixed magnetic field (Fig. 2d). Focusing on the conduction flat band, as the gate voltage increases from $V_{\text{gate}} \approx 2 \text{ V}$,

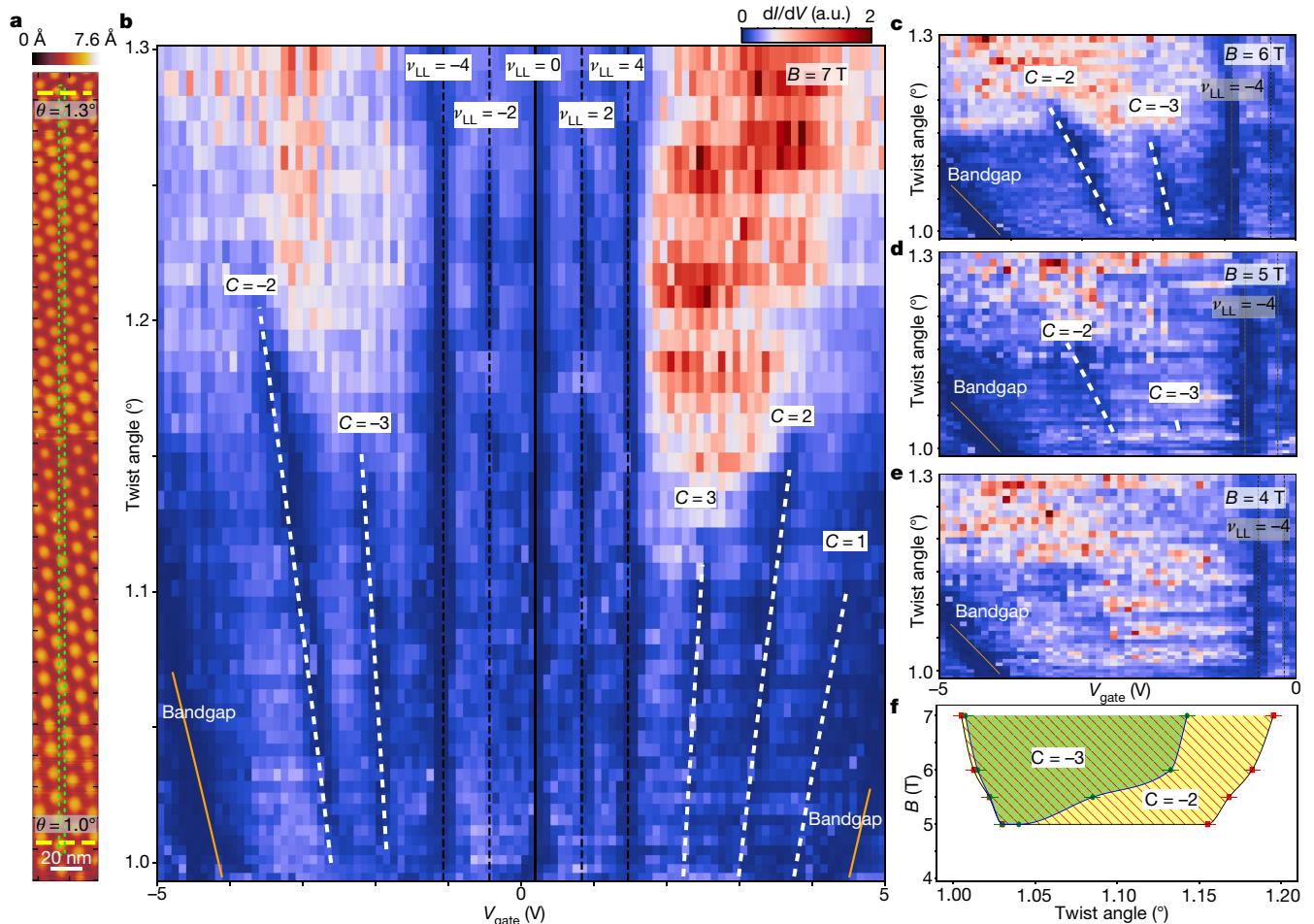


Fig. 3 | Angle and magnetic-field dependence of the LDOS and identification of Chern insulators. **a**, Topography of a 40 nm × 520 nm area where the twist angle gradually changes from 1.3° (top yellow line) to 1.0° (bottom yellow line); tunnelling conditions are $V_{\text{bias}} = 100$ mV, $I = 20$ pA. **b**, Conductance at $V_{\text{bias}} = 0$ mV and $B = 7$ T taken at different spatial points—characterized by different local twist angles θ —and for different V_{gate} values. Measurements were spatially averaged over the horizontal direction within the green dashed box shown in **a**. Chern insulating gaps develop only in a certain range of angles near the magic angle. The V_{gate} positions of the Chern insulating phases emanating from the

integer fillings shift with local twist angle θ , given that the non-zero filling depends on the moiré unit cell area, $A_m = \sqrt{3}L_m^2/2$. LL gaps originating from the CNP, by contrast, do not shift with V_{gate} . **c–e**, Magnetic-field dependence of the LDOS for hole doping ($-5 \text{ V} < V_{\text{gate}} < 0 \text{ V}$) at $B = 6$ T (**c**), $B = 5$ T (**d**) and $B = 4$ T (**e**). The Chern insulating gaps disappear as the magnetic field is lowered. **f**, Reconstructed phase diagram showing the range of fields and angles at which $C = -2$ and $C = -3$ Chern insulators are observed. In the shaded regions the suppression of the LDOS due to Chern insulators is prominent. See Extended Data Fig. 6 for high-resolution data resolving the $C = -1$ state.

the fourfold-degenerate VHS approaches the Fermi level. Just before crossing ($V_{\text{gate}} \approx 3$ V), a series of small gaps open up, accompanied by a set of nearly horizontal resonance peaks. These resonances are attributed to quantum dot formation around the tip due to the emergence of a fully gapped insulating state in its vicinity (see also Methods section ‘Quantum dot formation’). As V_{gate} increases further, part of the LDOS at the VHS position is abruptly pushed up in energy (seen at higher V_{bias} in Fig. 2d), reducing its spectral weight. Similar transitions are observed also near $V_{\text{gate}} \approx 4$ V and $V_{\text{gate}} \approx 5.1$ V, with the spectral weight reducing after each transition. This transition sequence is analogous to the $B = 0$ T cascade—although with the finite field, onsets shift to new V_{gate} positions that trail the location of nearby Chern insulating phases (see also the additional discussion in Methods section ‘Magnetic-field dependence of the cascade of phase transitions’ and Extended Data Fig. 3).

These observations can be explained within the Hofstadter picture (Fig. 1f, g), where each zLL carries a total Chern number of $C = +4$, and the VHS carries $C = -4$. Here the factors of four reflect spin–valley flavour degeneracy. When the VHS in the conduction band is empty, the Chern number of the occupied bands is $C = -4 + 4 \times 2 = +4$ (here, -4 comes from the VHS in the valence band, whereas $+4 \times 2$ comes from zLLs).

Consequently, the gap between the zLL and the bottom of the VHS follows the corresponding $C = +4$ slope in the LDOS Landau fan emanating from the CNP (marked with the grey line labelled $+4$ in Fig. 2c). As charge density increases, all four spin–valley flavours in the conduction band are populated equally, until the VHS reaches E_F and interactions underlying the cascade⁷ shift all carriers to a single-flavour VHS while the other three are pushed to higher energies (Fig. 2e). Because the added single-flavour VHS carries $C = -1$, the total Chern number is now $C = +3$, and the next corresponding gap in the LDOS Landau fan follows an accordingly reduced slope. This sequence repeats, creating a cascade.

Angle dependence

To verify the role of correlations on the Chern insulating phases, we extend the technique introduced in Fig. 2a–c to visualize the evolution of these phases with twist angle. For this purpose, we focus on an area where the angle slowly changes over hundreds of nanometres (many moiré periods) so that the twist angle is locally well defined and the strain is low ($< 0.3\%$) (Fig. 3a). By measuring the LDOS at E_F versus V_{gate} and spatial position, we image the development of the Chern phases

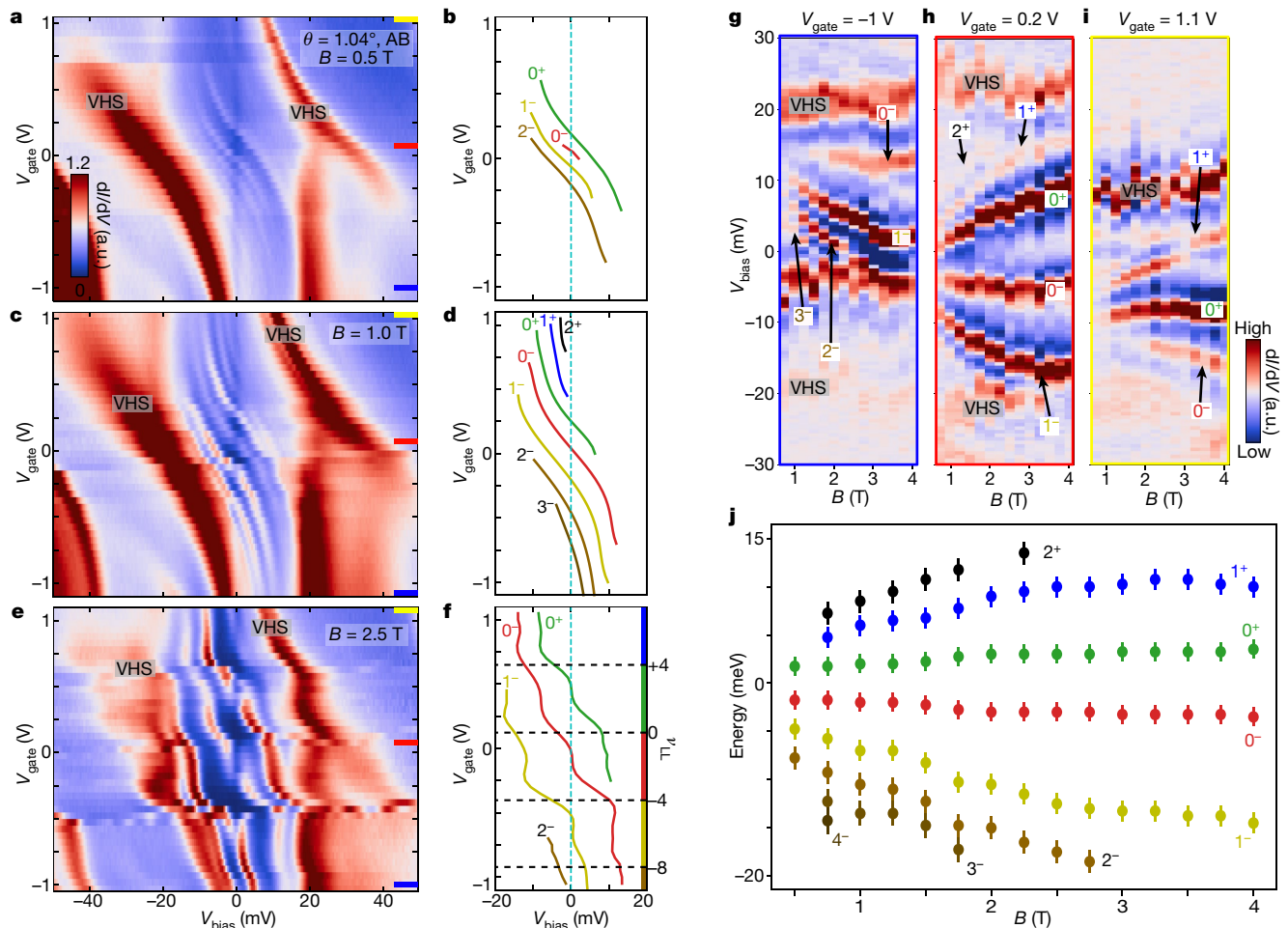


Fig. 4 | Evolution of LLs from charge neutrality. **a–f**, Point spectra as a function of V_{gate} (**a**, **c**, **e**) and schematics tracking the evolution of LLs (**b**, **d**, **f**) for $B = 0.5$ T (**a**, **b**), $B = 1$ T (**c**, **d**) and $B = 2.5$ T (**e**, **f**). At the Fermi energy, each LL is equally separated in electron density (V_{gate}), as marked by the LL filling factor ν_{LL} , indicating fourfold degeneracy of each level. **g–i**, Linecuts at $V_{\text{gate}} = -1$ V (**g**), $V_{\text{gate}} = 0.2$ V (**h**) and $V_{\text{gate}} = 1.1$ V (**i**) illustrating the LL spectrum change with

magnetic field. A smooth background has been subtracted to enhance visibility. The indicated LLs are identified in **b**, **d**, **f**, **j**. Combined energy spectrum for LLs around charge neutrality. Zero energy is set as the midpoint between the 0^+ and 0^- levels. Error bars in **j** are set by the measured width of the corresponding LLs.

and the zLLs (Fig. 3b). Whereas LL gaps around the CNP form at filling factors $\nu_{\text{LL}} = \pm 4, \pm 2, 0$ at fixed V_{gate} independent of twist angle, the Chern insulating phases move away from the CNP for larger angles, as expected from the change of the moiré unit cell size. In addition, the Chern insulating phases are observed only in a certain narrow range around the magic angle (Fig. 3b), highlighting the importance of correlations for their formation. For example, the $C = -3$ state emanating from $\nu = -1$ appears only for $1.02^\circ < \theta < 1.14^\circ$ at $B = 7$ T, whereas the $C = -2$ state is stable for a slightly larger range. Moreover, at lower fields, the angle range in which the Chern phases are visible also diminishes (Fig. 3c–e). The phase diagram (Fig. 3f) thus depends sensitively on both the magnetic field and the twist angle.

The observed evolution of Chern insulators with twist angle reflects a competition between Coulomb interactions and kinetic energy, similarly to the cascade at $B = 0$ T. Here the electron–electron interaction scale is approximately set by $U \approx (e^2/4\pi\epsilon L_m)$ (e and ϵ denote the electron charge and dielectric constant, respectively) and increases for larger twist angles. By contrast, the typical kinetic energy scale, taken to be the bandwidth W of the $C = -1$ single-flavour VHS, exhibits a non-monotonic dependence on the twist angle, is minimal at the magic angle, and further narrows with increasing magnetic field. We therefore expect Chern insulating phases to occur close to the magic angle, with larger fields required for their onset away from the magic

angle (Fig. 3f). Estimates of U/W as a function of B based on a continuum model support this reasoning (see Methods section ‘Theoretical modelling and comparison with experiment’). Further, we note a general electron–hole asymmetry in the angle range in which Chern phases are observed (Fig. 3b): the $C = -2$ phase emanating from $\nu = -2$ on the hole side is seen for $1.19^\circ > \theta > 1.01^\circ$, whereas the $C = 2$ state on the electron side is observed for $1.15^\circ > \theta$, down to a minimal angle of 0.99° accessible in this area. This observation indicates that the ‘magic angle’ condition for which correlations are strongest differs between the conduction and valence flat bands, and highlights the sensitivity of MATBG physics to tiny twist-angle changes.

Spectroscopy of LLs near the CNP

Here we explore low-magnetic-field LLs observed near the CNP, which are directly relevant for the MATBG band structure. Many MATBG properties remain poorly understood—for example, the appearance of fourfold-degenerate LLs around charge neutrality¹ instead of eightfold, as expected from the eight degenerate Dirac cones of the two stacked monolayers, as well as the measured flat-band bandwidth (~ 40 meV)^{19–22}, which far exceeds theoretical predictions¹⁰. These discrepancies are largely due to difficulties in performing band-structure calculations that incorporate electronic correlations^{30–32}, strain³³ and atomic

reconstruction³⁴. Several mechanisms have been proposed to explain the fourfold-degenerate LLs formed at charge neutrality^{33,35,36}, although no general consensus has emerged. Figure 4a–f shows the evolution of CNP LLs for $0.5 \text{ T} < B < 4 \text{ T}$. The fourfold LL degeneracy, revealed in Fig. 2c, is seen here through the equal separation between LLs at V_{gate} as they cross E_{F} (see v_{LL} in Fig. 4f and compare the colour coding from the diagram in Fig. 2c). Here, we focus on the relative energy separation between LLs taken at fixed V_{gate} while varying B (Fig. 4g–i). Because the energy separation between LLs can be affected by exchange interactions when LLs cross the Fermi energy, we choose V_{gate} values to avoid such interaction-altered regions and compile the single-particle energy spectrum, shown in Fig. 4j. We note that although fine features of the LL spectrum may vary with spatial location, the energy separations remain similar.

The observed LL spectrum is consistent with a scenario wherein the two moiré Brillouin zone Dirac cones are shifted in energy, either by strain (0.3% in this area)^{33,36} or by layer polarization due to a displacement field³⁷ (see also Methods section ‘LL spectrum for low fields and MATBG band structure’). In this scenario, 0^+ and 0^- (Fig. 4j) originate from the two Dirac points, and the spectrum can be compared to the Dirac-like dispersion $E_n = \text{sgn}(n)v_{\text{D}}\sqrt{2e\hbar|n|B}$, where $\text{sgn}(n) = n/|n|$ and $\hbar$ is the reduced Planck constant. The observed LL separations (for example, 8 meV for 0^- to 1^- and 5 meV for 1^- to 2^- at $B = 2 \text{ T}$) yield a Dirac velocity of $v_{\text{D}} \approx (1.5\text{--}2) \times 10^5 \text{ m s}^{-1}$, which exceeds continuum-model predictions by an order of magnitude, highlighting possible interaction-induced modification of MATBG bands near the CNP. Electron–hole asymmetry is also present, as the energy differences between the first few LLs on the hole side are larger than their electron-side counterparts. Moreover, upon doping, LLs move together towards the VHS, whereas the separation between them is hardly affected (Fig. 4a, c, e). This finding signals that dispersive pockets within the flat bands do not change substantially as flatter parts of the bands deform^{30,31}. Together these observations place strict restrictions on the MATBG electronic bands and provide guidance for further theoretical modelling. Looking ahead, we anticipate that the STM spectroscopic techniques developed here will enable the exploration of other exotic phases in MATBG and related moiré systems.

Note: in the course of preparation of this manuscript, we became aware of a related work³⁸.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-020-03159-7>.

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Methods

Device fabrication

The device used in this study was fabricated with the polydimethylsiloxane (PDMS)-assisted stack-and-flip technique developed in our previous work²⁰. Thick (~30 nm) hBN, monolayer WSe₂ (purchased from HQ Graphene) and monolayer graphene are exfoliated on SiO₂ and identified optically. We use a poly(bisphenol A carbonate) (PC)/PDMS stamp to pick up hBN and WSe₂, and use the stack to tear and twist graphene layers¹⁷, all at -90 °C. We manually rotate the two graphene pieces by -1.0°. The entire stack attached to the PC is then transferred onto a clean PDMS, with the twisted bilayer graphene side facing the PDMS. We dissolve PC in *N*-methyl-2-pyrrolidone and the stack is transferred onto a chip with a graphite gate and gold electrodes. In the final step, twisted bilayer graphene is connected to the electrodes by another graphite flake. This method results in good sample quality where we can easily find a contamination-free area greater than 400 nm × 400 nm (Extended Data Fig. 1b) and negligible surface corrugation due to the graphite back gate.

STM measurements

The STM measurements were performed in a Unisoku USM1300J STM/AFM system on STM mode. We used a platinum iridium (Pt/Ir) tip, first prepared on a Ag(111) crystal. The tip quality was verified by measuring quasiparticle interference and a spectrum that shows the Ag(111) surface state. Additionally, the tip was cleaned on the MATBG sample by pulsing on the gold electrode and verifying the resulting spectrum³⁹. It is likely that the final tip also contained Au atoms. All the features reported here are robust, obtained with more than ten different microtips. The measurements were performed at 2 K and the lock-in parameters were modulated at voltage $V_{\text{mod}} = 0.5\text{--}1\text{ mV}$ and frequency $f = 973\text{ Hz}$; the set parameters for dI/dV were $V_{\text{bias}} = 100\text{ mV}$ and $I = 1\text{ nA}$, unless specified otherwise. The piezo scanner was calibrated on a Pb(110) crystal. The twist-angle uncertainty in the MATBG measurements is $\pm 0.01^\circ$, originating from measured lengths in topography.

The LDOS Landau fan measurements shown in Fig. 2c were performed by taking STM data in the feedback-off mode while sweeping V_{gate} . After the dataset was collected, we changed the magnetic field while keeping track of the same spatial position and turning the feedback on. For each magnetic field, the LDOS data at the Fermi energy were normalized by the average value.

Effects of WSe₂

One of the main differences between this work and previous MATBG results is the use of monolayer WSe₂ as a substrate. This approach enables finding larger areas (400 nm × 400 nm) that are cleaner and have much less twist-angle disorder. Apart from this technical difference, the effects of spin-orbit interaction (SOI), studied in ref. ¹⁷, have corresponding energy scales of 1 meV—similar to the LL broadening observed with STM at 2 K. For this reason, we believe that SOI effects do not play a critical role in the results reported here. Also, the features observed in zero field qualitatively match results obtained in MATBG on hBN⁸. However, it is plausible that some WSe₂-related effects⁴⁰ combined with different measurement geometry, such as possible enhancement of Rashba SOI, may affect details of the MATBG band structure. Future studies will be focused on a systematic investigation of MATBG placed on different substrates.

According to optical inspection (Extended Data Fig. 1a), WSe₂ and MATBG are not aligned. Also, because WSe₂ and graphene have very different lattice constants, in contrast to hBN, global C₂ symmetry breaking is unlikely, even in the case of perfect alignment. Importantly, our STM measurements do not show any C₂ symmetry-breaking signatures in electronic structure, and we find that signatures in topography of the possible moiré pattern between MATBG and WSe₂ are much weaker compared to similar signatures that can be observed in MATBG/

hBN structures. This is not surprising, because the length scale of the MATBG/WSe₂ moiré pattern is expected to be around 1 nm, thus comparable to inter-atomic distances.

Filling-factor assignment

From the LDOS Landau fan diagram (Fig. 2c), we can precisely identify gate voltages that correspond to the origins of different LLs and Chern insulator states. Once gate voltages corresponding to $\nu = 0$ and $\nu = \pm 4$ (sometimes also ± 2) are identified, conversion between gate voltage and filling factor is fully defined. We have verified that this conversion also gives twist angles that match the values estimated from topography. This approach, equivalent to the one previously used in transport measurements, is much more precise than the alternative methods used in all previous MATBG STM measurements^{8,19,20,22,38}, where the LDOS fan diagram is missing and the filling-factor assignments are more ambiguous; see Extended Data Fig. 10. By using the LDOS Landau fan, the full filling position can be determined with an error of less than about 0.1 V or, equivalently, a filling-factor error of 0.07 electrons per moiré unit cell. This is a factor of 3–5 smaller than the error obtained when determining the filling factor from the half-filling positions at zero magnetic field²⁰.

Quantum dot formation

In our high-resolution spectra, various gaps at the Fermi energy are accompanied by nearly horizontal resonance peaks and Coulomb diamonds. For example, bandgaps between the remote bands and the flat band (Extended Data Fig. 5a at $V_{\text{gate}} \approx \pm 6\text{ V}$), gaps between LLs (Fig. 4e, Extended Data Fig. 4b) and Chern insulating gaps (Fig. 2d) all show resonances around the gaps. We note that the majority of resonances and Coulomb diamonds are formed below (in V_{gate}) the gap position expected from the LDOS Landau fan. This observation can be explained by quantum dot formation around the tip. Because the STM tip and graphene have different work functions²⁰, even at zero bias the tip acts as a top gate and slightly dopes the region in the immediate vicinity. The likely presence of Au atoms on the tip and the observed positions of resonance peaks are both consistent with tip-induced hole doping underneath. Consequently, as we increase V_{gate} , the global sample enters the insulating regime first, and the area underneath the tip can still be conducting—thereby creating a conducting island (that is, quantum dot) and consequent resonances. Further doping will bring the area underneath the tip to the gapped phase, explaining that resonances are below the gap in V_{gate} . We interpret the resonances and observed Coulomb diamonds as direct evidence of fully developed gaps, given that quantum dots form only when the insulating-phase resistance exceeds the resistance quantum. Owing to the presence of quantum dots, extracting exact values of Chern insulating gaps is not straightforward, given that both the Chern gaps and the quantum dot charging energy (typically 2–3 meV) are comparable.

Magnetic-field dependence of the cascade of phase transitions

Extended Data Figure 3 shows point spectra at magnetic fields of $B = 4\text{--}8\text{ T}$. The positions of the dashed lines are calculated from the slopes in the LDOS Landau fan diagram with the appropriate Chern numbers. As discussed in the main text, for the conduction band, the onsets of the cascade of transitions, where the LDOS is pushed up away from the Fermi energy, match well with the positions expected from the Chern number and the magnetic field. For the valence band, the onsets of the cascade do not change until $B = 6\text{ T}$, where the correlated Chern insulating phases start to occur. For $B > 6\text{ T}$, the cascade onsets also match with the Chern number lines. We note that with or without a magnetic field, cascade features are more visible in the conduction band, because the LDOS at the Fermi energy becomes very low after crossing each cascade transition. In the valence band, however, the LDOS at the Fermi energy is not sufficiently suppressed after the transition. For example, around $\nu \approx -1$ the VHS peak is still pinned to E_F while

a small amount of spectral weight is pushed away; by contrast, at $\nu \approx 1$ the whole VHS peak is lifted from E_F . This observation again illustrates the electron–hole asymmetry present in MATBG and is consistent with our observation in the LDOS Landau fan (Fig. 2c) that LLs originating from the CNP may still be visible beyond $\nu = -1$ filling.

Magnetic-field dependence of flat-band LLs

LLs and zero-field features near the CNP. As shown in Fig. 4, in our experiment we can resolve LLs around the CNP down to $B = 0.5$ T (measured at $T = 2$ K). This resolution is in line with transport measurements taken at similar temperatures. However, even at zero magnetic field, we often see small additional LDOS peaks around the CNP within the VHS (see Fig. 1c); we note that their intensity is much lower compared to the main LDOS peaks (VHSs) that are determined by the flat parts of the band structure. Mostly probably, these minor peak features at $B = 0$ T are related to MATBG band-structure details that are not captured by simple continuum models. Recent *ab initio* modelling suggested that the twisted bilayer graphene moiré Brillouin zone can be considerably distorted, with local minima/maxima residing close to the Dirac points³¹. These local extrema in the band structure would manifest as small peaks in the LDOS observed by STM. We also note that similar fine LDOS peaks at $B = 0$ T were recently observed in a very different quantum material with Dirac dispersion⁴¹, possibly indicating that other effects, such as local fluctuations in chemical potential, can also play a role. Importantly, at $B > 0.5$ T the LLs become more prominent and can be easily distinguished from these zero-field features by their systematic shift in magnetic field and V_{gate} .

Quantum Hall ferromagnetism of 0^+ and 0^- LLs. When partially filled, the two fourfold-degenerate LLs closest to the CNP, 0^+ and 0^- , exhibit quantum Hall ferromagnetism at high magnetic fields (Extended Data Fig. 4b, c). Consider the evolution of the 0^+ level. Starting from the top linecut in Extended Data Fig. 4c, as V_{gate} is lowered, the 0^+ level splits several times when crossing the Fermi energy owing to the exchange interaction. Each time, only the part of the LDOS corresponding to one of the flavours is transferred above E_F . By tracing the evolution of 0^+ , we can resolve splitting between all four flavours. We note, however, that the visibility of such flavour-resolved LLs (that is, the corresponding LDOS) is not equal, indicating that the filled LLs in the quantum Hall ferromagnetic states have different spatial distribution. In particular, the dI/dV peak at $V_{\text{bias}} < 0$ mV labelled 2 in the upper part of Extended Data Fig. 4c is of similar height as the one above (labelled 3), whereas the last peak (labelled 1) is quite reduced, that is, the LDOS there is low. We observe similar LDOS evolution of the 0^- level. These variations possibly originate from layer polarization. That is, some of the flavour-resolved LL wavefunctions may live predominantly on the bottom layer—thereby reducing the corresponding LDOS measured by the STM tip on the top layer. These effects can also reduce the visibility of different ν_{LL} gaps in Figs. 2, 3, where odd ν_{LL} are difficult to resolve ($\nu_{\text{LL}} = +1$ is ‘missing’ in Fig. 2c and $\nu_{\text{LL}} = \pm 1, \pm 3$ in Fig. 3; see also Extended Data Fig. 6).

Electron–hole asymmetry. Our STM observations highlight electron–hole asymmetry visible in the LL evolution of MATBG. We note that measurements showing this asymmetry in Figs. 2, 3 differ fundamentally from the ones discussed in Fig. 4. Figures 2c, 3 present gaps at the Fermi level that develop owing to formation of LLs or Chern insulators and access physics close to E_F ; in this context, these measurements are similar to transport. The electron–hole asymmetry of MATBG in these measurements is inherited from the gap evolution with magnetic field and V_{gate} . For example, Fig. 3—which shows the gap evolution with twist angle—suggests that flat portions of the valence flat band that make up the VHS evolve differently from the corresponding conduction band. Concretely, starting from angles above the magic angle, the bandwidth of the valence band narrows more rapidly than the conduction band and becomes sufficiently narrow for the development of Chern insulators at

larger twist angles. As the twist angle is further lowered, the bandwidth increases again. By contrast, the bandwidth of the conduction flat band becomes sufficiently narrow to give rise to Chern insulators at lower twist angles (around 1.14° at $B = 7$ T). Ultimately, at around 1° , our data are consistent with the flat portion of the conduction band becoming narrower than the valence band, given that Chern insulators become visible at lower magnetic fields on the electron side; recall Fig. 2c.

In contrast to this observation of MATBG Fermi-level physics, Fig. 4 shows the LL energy spectrum when the LLs are far from E_F and thus reflects directly the asymmetry in the shape of MATBG flat bands near the Dirac points.

Theoretical modelling and comparison with experiment

Chern number assignment for MATBG in a magnetic field. Here we discuss the assignment of Chern numbers to the MATBG magnetic-field band structure shown in Fig. 1g. To understand the Chern number assignment, it is helpful to start from monolayer graphene. When placed in a magnetic field⁴², graphene’s quantum Hall conductivity per spin is quantized in steps of $(2n + 1)e^2/h$, where n is the LL index as defined in the main text, e is the elementary charge and h is the Planck constant. By relating this with the conventional formula for the quantum anomalous Hall conductance per spin, Ce^2/h , we find that the 0th LL has a Chern number of $C = 1$ (the sign of C and its relation to the quantum anomalous Hall conductance is a convention; we could equally well have specified $C = -1$). With each consecutive LL, the total Chern number of occupied states increases until the VHS, when it starts to fall, with the conductance of the two filled bands vanishing⁴³. This reasoning holds also for MATBG, where the zLL similarly carries Chern number $C = 1$ (ref. ³⁵). However, as discussed in the main text, because of how flat the original bands are, in high fields it is experimentally difficult to resolve other flat-band LLs beyond the zLL. It is therefore helpful to think of the zLLs individually, and group the remaining LLs into two sub-bands centred around the electron/hole VHS. By doing so, we can then assign Chern number $C = -1$ to the two ‘effective’ sub-bands containing the VHS, given that the overall two bands, when fully filled or empty, must have vanishing Hall conductance (and hence total Chern number zero). In the main text we refer to this VHS-centred sub-band as simply ‘VHS’.

Modelling. We used both the continuum and ten-band models to calculate the energy spectrum of MATBG subjected to magnetic fields (see Supplementary Information for calculation details). The continuum model is known to qualitatively capture the correct physics of twisted bilayer graphene at twist angles far from the magic angle. However, it underestimates the experimentally observed bandwidth energy scale in MATBG by an order of magnitude²⁰, and thus does not capture the correct low-field LL spectrum near the CNP. This discrepancy is in part expected because strong correlations, alongside atomic reconstruction, play a critical role in the physics of MATBG. For this reason, our analysis also employed the phenomenological ten-band model, which keeps track of the MATBG symmetries while also possessing a range of parameters that can be easily tuned to allow a more direct comparison with the experiment. Results from the continuum model are presented in Extended Data Fig. 7. The ten-band model results, discussed in the context of the low-energy LL spectrum, are shown in Extended Data Fig. 9.

U/W evolution with magnetic field and phase diagram of Chern insulating phases

Extended Data Fig. 8 shows an estimate of the ratio between the characteristic interaction energy U and kinetic energy W as a function of twist angle θ and magnetic field B . Results in Extended Data Fig. 8a, b (Extended Data Fig. 8c, d) follow from a continuum model (see Supplementary Information for calculation details). The bandwidth W is chosen as the width of the $C = -1$ Chern bands that includes the VHS but not the zLLs; we denote the bandwidths on the electron and hole

sides by W_c and W_h , respectively (see Extended Data Fig. 7b). The characteristic interaction scale, as specified in the main text, is taken to be $U = e^2/(4\pi\epsilon\epsilon_0)$. Roughly, beyond a critical U/W , we expect cascade transitions to become energetically favourable, leading to the experimentally observed Chern insulator phases. Extended Data Fig. 8a, b (Extended Data Fig. 8c, d) depicts constant U/W contours (taking $\epsilon = 5\epsilon_0$, where ϵ_0 is vacuum permittivity) for various twist angles, excluding a narrow window close to the magic angle where the bandwidths cannot be reliably extracted. These constant U/W contours qualitatively reproduce trends observed in the experiment (Fig. 3f), most notably the minimum of the U/W contours near the magic angle. The model's small but non-zero electron-hole asymmetry—manifest in the relative shift of contours in Extended Data Fig. 8a, b (Extended Data Fig. 8c, d)—is also consistent with observations.

The main trends for U/W evolution with magnetic field and twist angle can be captured analytically, thus further elucidating the underlying physics. To proceed, we estimate the dependence of the bandwidth W on magnetic field and twist angle. As shown in Extended Data Fig. 7, each $C = -1$ band narrows with field from its $B = 0$ value W_0 (which is half of the full zero-field flat-band bandwidth). We crudely capture this narrowing as follows. First, the inner $C = -1$ band edges, close to the CNP, are modified by LLs descending from charge neutrality with energies $\propto \sqrt{B}$. The outer $C = -1$ band edges in general further squeeze the bandwidth by a model-dependent quantity. For simplicity, however, we take the bandwidth reduction arising from the upper and lower edges of a given $C = -1$ band to be identical. This approximation yields an effective bandwidth of

$$W(B) \approx W_0 - 2\nu_F \sqrt{2eB}, \quad (1)$$

where ν_F is the twist-angle-dependent velocity of the Dirac cone, and the factor of 2 comes from our rough assumption of symmetric band squeezing ($\hbar = 1$ in this section). The critical field B_c at which Chern insulators appear, corresponding to the edges of the regions in Fig. 3f, correspond to a solution of $U/W(B_c) = \eta$ (with $\eta > 1$), that is,

$$B_c \approx \frac{W_0^2}{8e\nu_F^2} \left(1 - \frac{1}{\eta} \frac{U}{W_0} \right)^2. \quad (2)$$

We estimate the $B = 0$ bandwidth as set by $W_0 \approx \nu_F \kappa$, where $\kappa = 4\pi/(3L_m)$ is the κ point at the corner of the mini-Brillouin zone. Finally, to find a simple expression for ν_F as a function of twist angle, we follow ref.¹⁰ and write

$$\begin{aligned} \frac{\nu_F^2}{\nu_0^2} &= \left(\frac{1 - 3\alpha^2}{1 + 6\alpha^2} \right)^2 + \delta\nu_F^2, \\ \alpha &= \frac{w_1}{\nu_0 \kappa}, \end{aligned} \quad (3)$$

with $\nu_0 = 10^6 \text{ m s}^{-1}$ the monolayer graphene band velocity and w_1 the moiré interlayer coupling parameter that defines the magic angle. As required¹⁰, at the magic angle the term in parenthesis above vanishes. To allow for a finite bandwidth, we have included a broadening parameter $\delta\nu_F$. We plot equation (2) for several values of η in Extended Data Fig. 8e. In the plot we use $\delta\nu_F = 0.013$, $w_1 = 117 \text{ meV}$ and $\epsilon = 5\epsilon_0$. The essential feature of the experimental phase diagram—that is, the minimum near the magic angle—is correctly reproduced. The observed phase boundary, as reproduced by the numerical estimates of Extended Data Fig. 8a, b (Extended Data Fig. 8c, d) is however much steeper with magnetic field—a feature not captured by the crude approximation for $W(B)$ introduced in equation (1). The above analytical and numerical analyses demonstrate the critical roles of the twist angle and magnetic field in forming the Chern insulating states driven by correlations.

LL spectrum for low fields and MATBG band structure

As discussed in the main text, our LL spectrum, shown in Fig. 4j, provides essential input about MATBG band structure. Besides setting a lower bound on the Dirac velocity through the spacing of LLs, it indicates the likely presence of gapless (or nearly gapless) Dirac cones in the MATBG band structure, contrasting some recent theoretical studies. Here we provide a more detailed analysis of different MATBG band-structure scenarios near the CNP. For this purpose, we use the ten-band model and the corresponding Hofstadter spectra to qualitatively match the experimental findings. Extended Data Fig. 9a–h provides an overview of the different band-structure scenarios that we considered. Throughout, we stipulate that the Dirac points either remain fixed to the κ, κ' points or are gapped, although in practice they may be also shifted by the presence of strain³³. Such shifts, however, do not have a substantial impact on the energy spectrum of the system or our analysis. In the following, we briefly outline four possibilities for the MATBG band structure.

(i) Dirac cones at κ, κ' are strongly gapped by the presence of strong correlations, and the dispersion around the centre of the moiré Brillouin zone (MBZ) is inverted, as proposed by several recent theoretical studies³² (Extended Data Fig. 9a, b). This scenario implies that all LLs originate from states near the MBZ centre, which would require an unrealistic linear-like dispersion to accommodate our observed LL structure—most notably, nearly flat zLLs. It may also imply a fully gapped system around the CNP even at zero field (unless the flat bands touch at the MBZ centre or exhibit some other dramatic reconstruction), which is not consistent with our observations. In our simple simulations, even when the bands do touch near the MBZ centre, the separation between zLLs increases much more rapidly than observed in experiments.

(ii) Dirac cones at κ, κ' remain gapless and the parts of flat bands near the MBZ centre merge with the VHS. In this scenario, all observed LLs originate from the Dirac dispersion (Extended Data Fig. 9c, d). Energetically this scenario is consistent with the observed spectrum and reproduces all the relevant features, including the $\sqrt{B}$ dependence of the LL spacing. However, we stress that in this scenario, likely accidental crossings between LLs originating from different Dirac cones have not been observed. A possible explanation for the absence of these crossings involves Dirac cones being highly asymmetric, with band curvature leading to different Dirac velocities for electron-like and hole-like excitations. Curvature effects, combined with the fact that the energetically shifted Dirac points always reside closer to one of the VHS than the other, may explain the absence of certain LLs. That is, the 'missing' LLs could simply be absorbed into the nearest VHS for low fields, before they become visible in experiment (-0.5 – 0.75 T).

(iii) Dirac cones at κ, κ' remain gapless, and the dispersion around the MBZ centre becomes heavily inverted—giving rise to a subset of the observed LLs (Extended Data Fig. 9e, f). For example, within this scenario, the Dirac points can contribute the 0^+ and 0^- levels, whereas all LLs originate from the inverted band structure near the centre of the moiré Brillouin zone. In this explanation, the relative separation between 0^- and 1^- (and between 0^+ and 1^+) would not be determined by the Dirac velocity, but by band-structure details (such as the relative energy difference between the Dirac point and the pockets around the MBZ centre, as well as the shape of the pocket near the MBZ centre). Viability of this scenario requires the Dirac velocity ν_D to exceed $(2.5\text{--}3) \times 10^5 \text{ m s}^{-1}$; otherwise, additional LLs descending from the Dirac cones would be visible.

(iv) One Dirac cone is gapped and the other merges with a VHS (Extended Data Fig. 9g, h). This scenario is outlined only for completeness, as it can be easily ruled out on the basis of the main features observed in the Hofstadter spectrum. In general, the crossing between the zLL and other levels (originating from the other Dirac cone) must occur at a certain finite field, which was not observed experimentally.

Based on this analysis, we believe that only scenarios (ii) and (iii) are plausible for MATBG band structure. Further research is needed to fully

distinguish between them. We also note that a small mass term may result in zLLs that disperse with the field. As shown in Fig. 4, we observed a mild increase in the separation between 0^+ and 0^- levels as the field increased. This feature is not universal, however, as occasionally these levels would move in parallel or approach each other, consistent with the existence of a non-spatially uniform mass term that determined the evolution of these levels with the field. Future work will be focused on relating the LL evolution to possible strain effects.

Data availability

The data that support the findings of this study are available from the corresponding authors on reasonable request.

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Acknowledgements We acknowledge discussions with A. Young, G. Refael and S. Mashhadi. The device nanofabrication was performed at the Kavli Nanoscience Institute (KNI) at Caltech. This work was supported by NSF through grants DMR-2005129 and DMR-1723367 and by the Army Research Office under grant award W911NF-17-1-0323. Part of the initial STM characterization was supported by CAREER DMR-1753306. Nanofabrication performed by Y.Z. was supported by the DOE-QIS programme (DE-SC0019166). J.A. and S.N.-P. also acknowledge the support of IQIM (an NSF Physics Frontiers Center with support of the Gordon and Betty Moore Foundation through grant GBMF1250). Y.P. acknowledges support from a startup fund from California State University, Northridge. A.T., C.L. and J.A. are grateful for support from the Walter Burke Institute for Theoretical Physics at Caltech and the Gordon and Betty Moore Foundation's EPIQS Initiative, grant GBMF8682. Y.C. and H.K. acknowledge support from the Kwanjeong fellowship.

Author contributions Y.C. and H.K. fabricated samples with the help of R.P., Y.Z. and H.A., and performed STM measurements. Y.C., H.K. and S.N.-P. analysed the data. Y.P. and A.T. implemented the models. Y.P., A.T. and C.L. provided theoretical analysis, supervised by J.A.; K.W. and T.T. provided materials (hBN). S.N.-P. supervised the project. Y.C., H.K., Y.P., A.T., C.L., J.A. and S.N.-P. wrote the manuscript.

Competing interests The authors declare no competing interests.

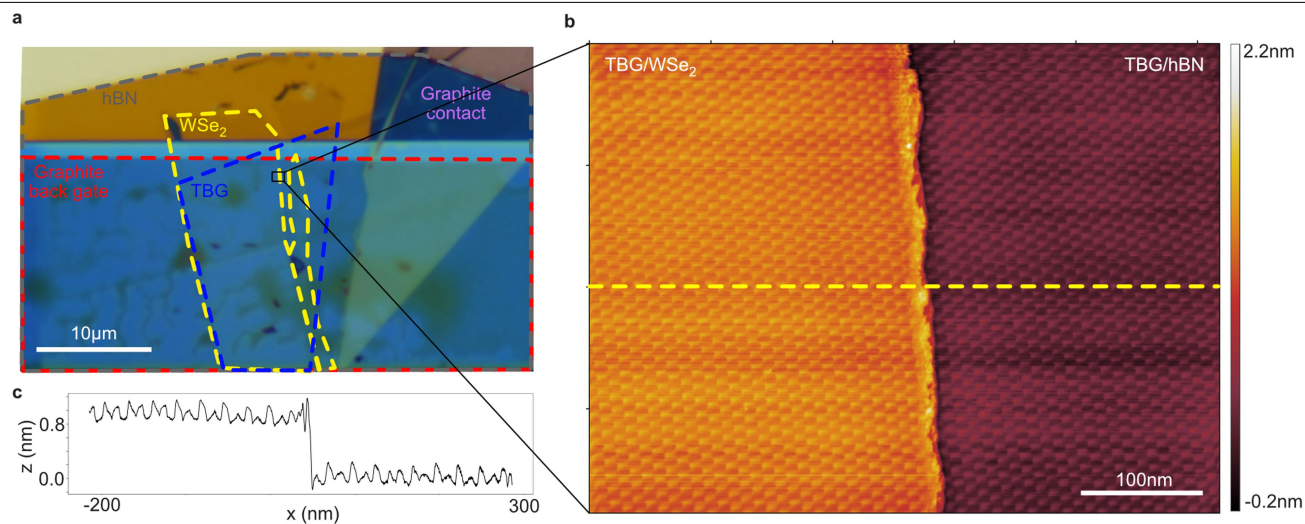
Additional information

Supplementary information is available for this paper at <https://doi.org/10.1038/s41586-020-03159-7>.

Correspondence and requests for materials should be addressed to S.N.-P.

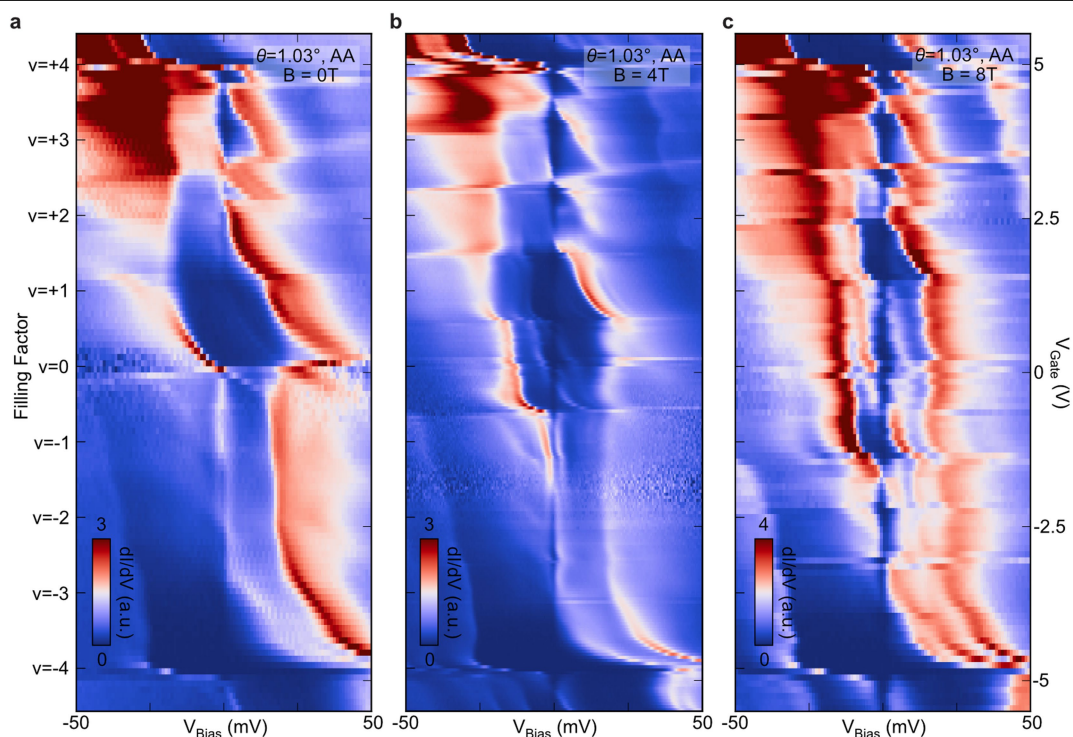
Peer review information Nature thanks Vincent Renard, Yayu Wang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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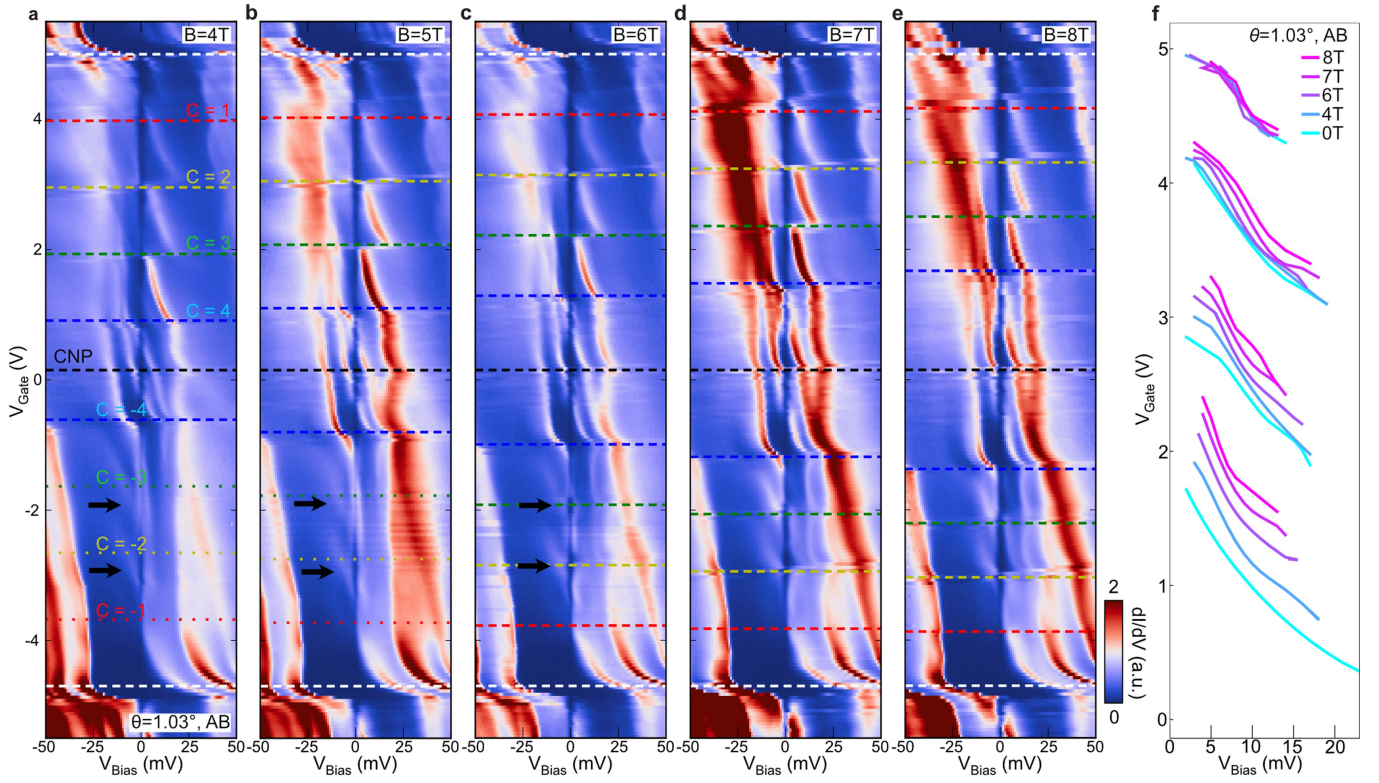


Extended Data Fig. 1 | Optical image of the device and STM topography of WSe₂/hBN boundary. **a**, Optical image of the device. The blue dashed line indicates the region with twisted bilayer graphene (TBG). **b**, STM topography of a WSe₂ boundary underneath twisted bilayer graphene ($V_{\text{bias}} = -400$ mV,

$I = 20$ pA). **c**, Linecut along the yellow dashed line in **b**. The height difference is around 0.8 nm, which matches the monolayer WSe₂ thickness. Small ripples originate from the moiré pattern.

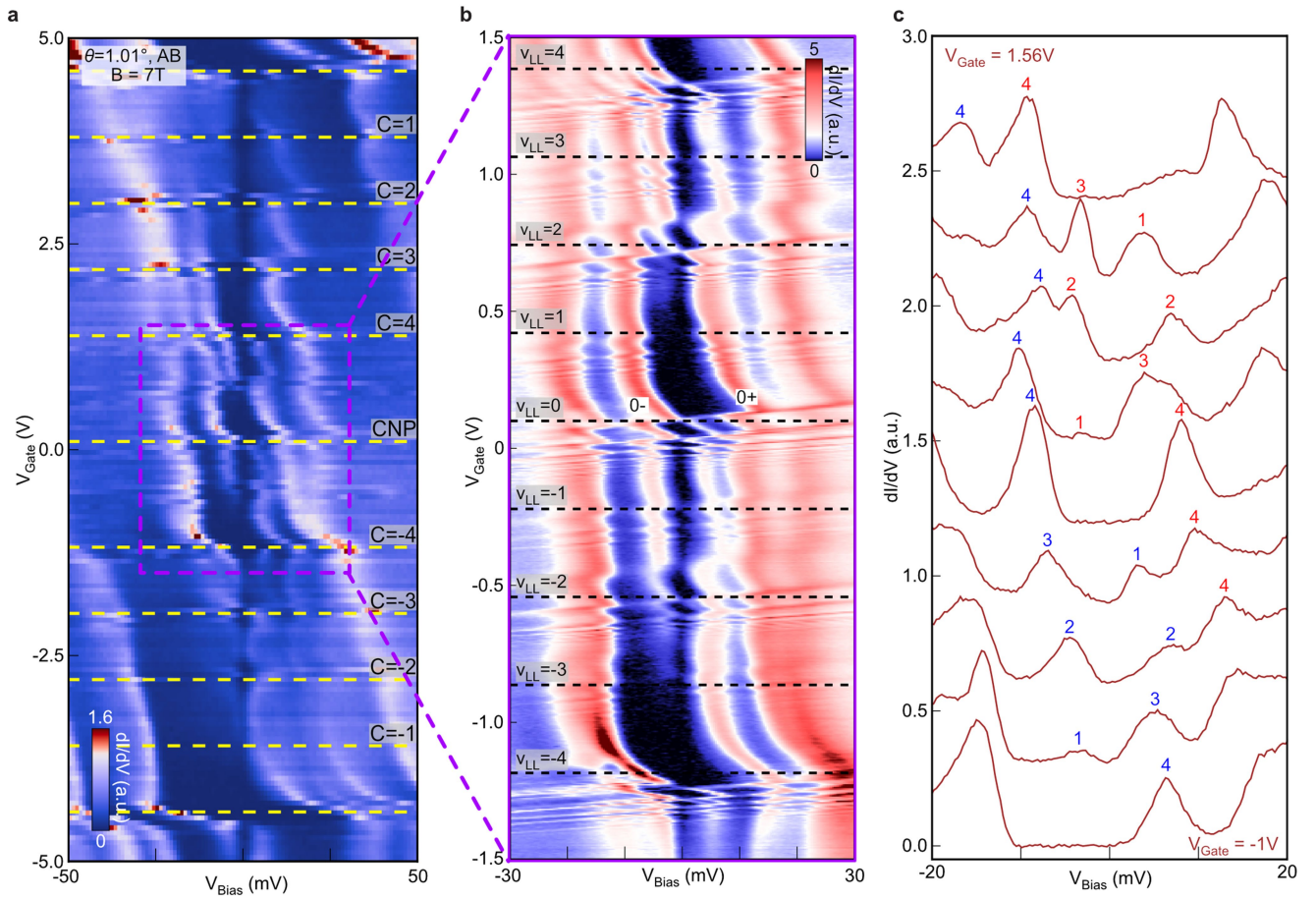


Extended Data Fig. 2 | AA site point spectra versus gate voltage with magnetic field. a, $B = 0$ T. b, $B = 4$ T. c, $B = 8$ T. The area (with $\theta = 1.03^\circ$) is the same as that shown in Fig. 1c–e. The VHSs are more pronounced compared to the AB site, making LL features harder to resolve.



Extended Data Fig. 3 | Evolution of cascade with magnetic field. **a–e**, Point spectra versus gate voltage on an AB site ($\theta = 1.03^\circ$) at $B = 4\text{ T}$ (**a**), $B = 5\text{ T}$ (**b**), $B = 6\text{ T}$ (**c**), $B = 7\text{ T}$ (**d**) and $B = 8\text{ T}$ (**e**). Coloured dashed lines indicate positions calculated with the corresponding Chern numbers. **f**, Tracking of cascades on

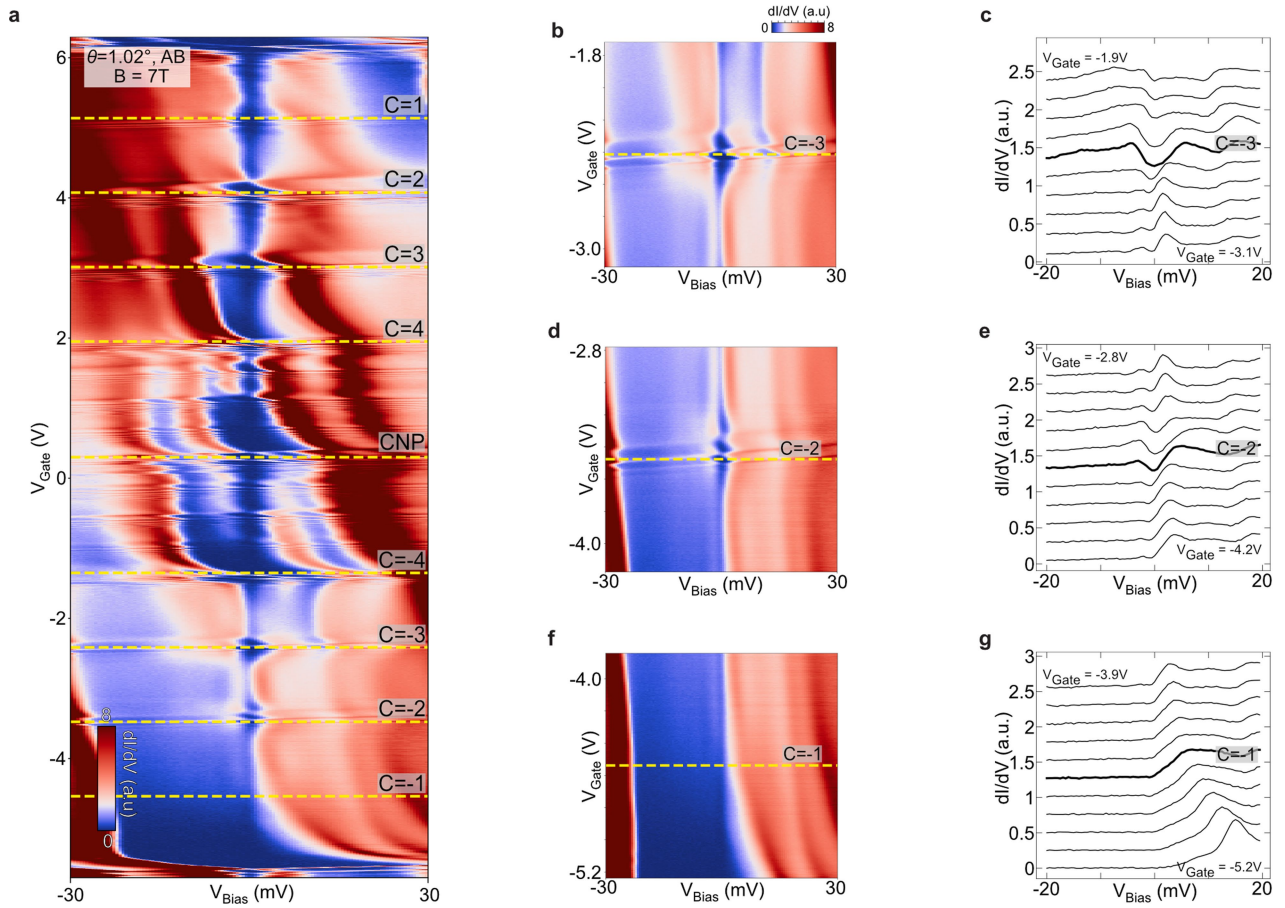
the conduction band at different magnetic fields. As the field is increased, the shifting of cascade onsets follows the evolution of Chern insulators. **a** and **e** show the same data as Fig. 1d, e; **b–d** are at intermediate fields on the same area.



Extended Data Fig. 4 | Quantum Hall ferromagnetic states at the zLL.

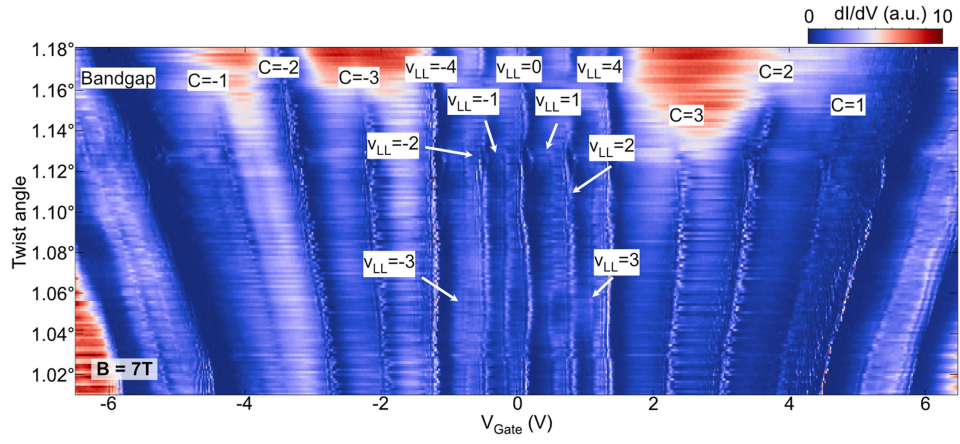
a, Point spectra on an AB site at $\theta = 1.01^\circ$ as a function of V_{gate} for magnetic field $B = 7$ T. **b**, Zoom-in around the CNP, where zLLs (0^- and 0^+) cross the Fermi energy. The fourfold degeneracy of each zLL is broken by exchange interactions at the Fermi energy, creating gaps at $\nu_{\text{LL}} = -4, -3, \dots, +4$. **c**, Linecuts

obtained when the 0^- and 0^+ LLs cross the Fermi energy, with vertical offset for clarity. Shifting of the spectral weight of the fourfold-degenerate 0^- and 0^+ LLs cross the Fermi level is clearly visible. Numbers in blue (red) correspond to the number of spin-valley flavours of 0^- (0^+) LLs below and above the Fermi level ($V_{\text{bias}} = 0$) as they cross E_F near $\nu_{\text{LL}} = 4, 3, 2, 1, 0, -1, -2, -3, -4$.



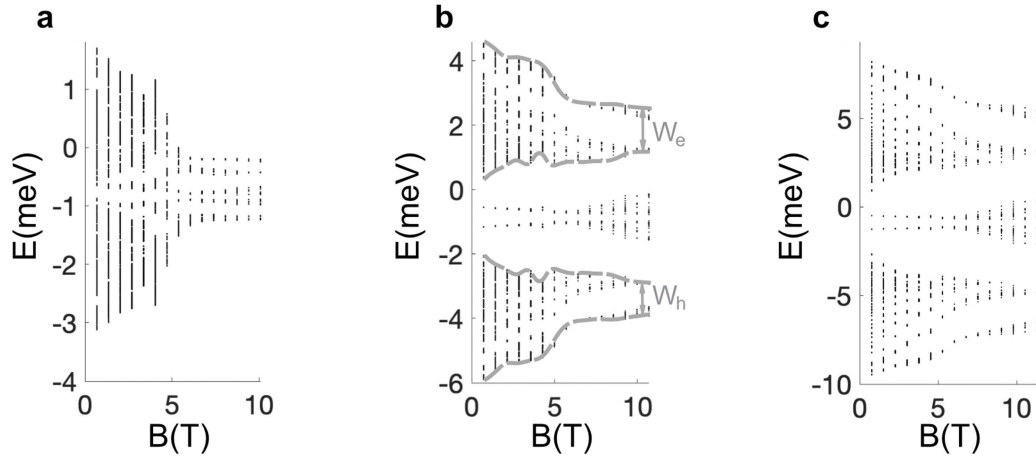
Extended Data Fig. 5 | Chern insulating phases in the valence band. **a**, Full gate voltage range of Fig. 2d at $B = 7$ T, from $\nu = -4$ to $\nu = 4$. **b–g**, Zoom-in of Extended Data Fig. 5a around $C = -3$ (**b**), $C = -2$ (**d**) and $C = -1$ (**f**), and the corresponding linecuts (**c**, **e**, **g**) at various V_{gate} values with vertical offset.

Shifting of the spectral weight across the Fermi level is clearly visible in **c** and **e**. The $C = -1$ state is not visible at this angle ($\theta = 1.02^\circ$) and this magnetic field ($B = 7$ T).



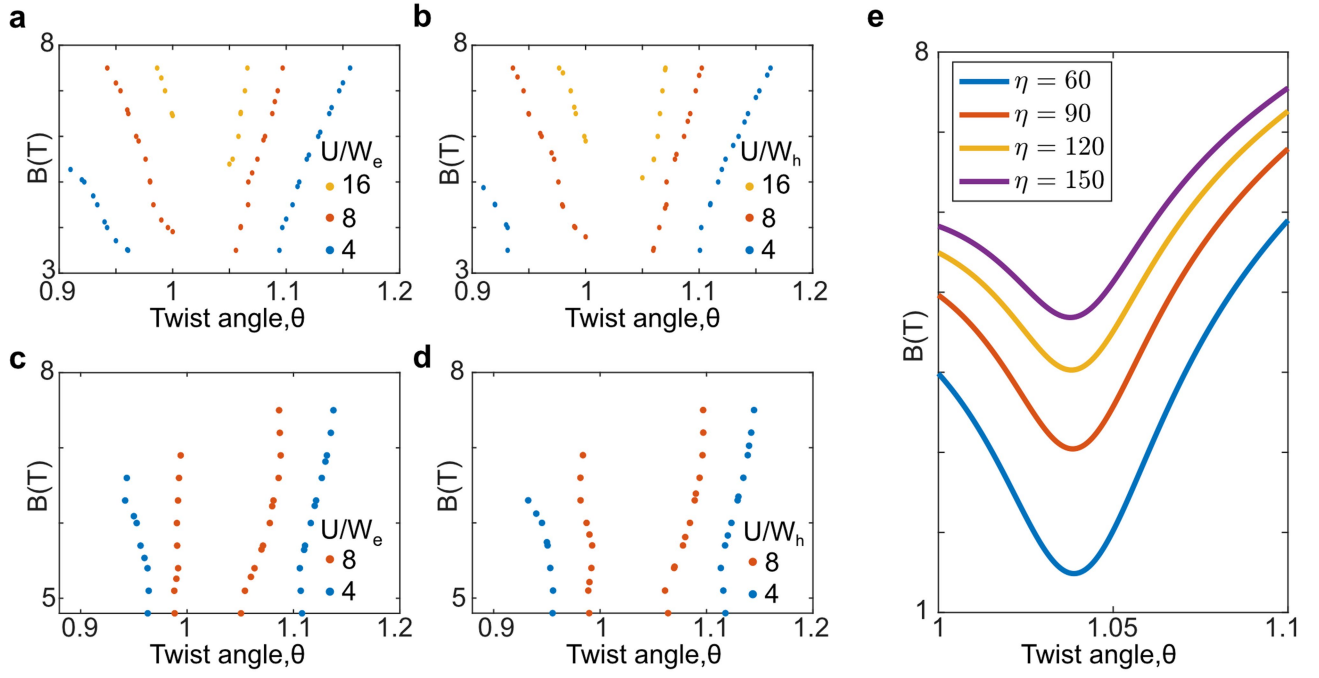
Extended Data Fig. 6 | $C = -1$ phase in a high-resolution angle-dependent LDOS map at $B = 7$ T. Measurements were taken over a $20 \text{ nm} \times 200 \text{ nm}$ area in which the twist angle varies continuously from $\theta = 1.01^\circ$ to 1.18° by using $V_{\text{bias}} = 0 \text{ mV}$ and changing the spatial position and carrier density (V_{gate}). The signature of the $C = -1$ phase and $\nu_{\text{LL}} = \pm 1, \pm 3$, which are not visible in Fig. 3b, are

now observed owing to the enhanced V_{gate} resolution, and the electron-hole asymmetry between Chern phases is even more pronounced. We note that the visibility of $\nu_{\text{LL}} = \pm 1, \pm 3$ shows strong spatial dependence, whereas the visibility of $\nu_{\text{LL}} = \pm 2, \pm 4$ is high throughout the range.



Extended Data Fig. 7 | Spectrum of a continuum model taking into account the effects of a displacement field. a–c, The spectra correspond to twist angles of $\theta = 1.03^\circ$ (a), $\theta = 1.06^\circ$ (b) and $\theta = 1.09^\circ$ (c). The displacement field, expressed as an energy difference between the two graphene monolayers, is 10 meV, and the ratio between AA and AB tunnelling amplitudes, $w_0/w_1 = 0.4$, is the same for all results. A similar spectrum is obtained for larger w_0/w_1 ratios. The simple continuum model, known to predict correct trends, can capture energetically offset Dirac cones by including a displacement field. However, it underestimates the bandwidth of the flat bands (or equivalently the separation

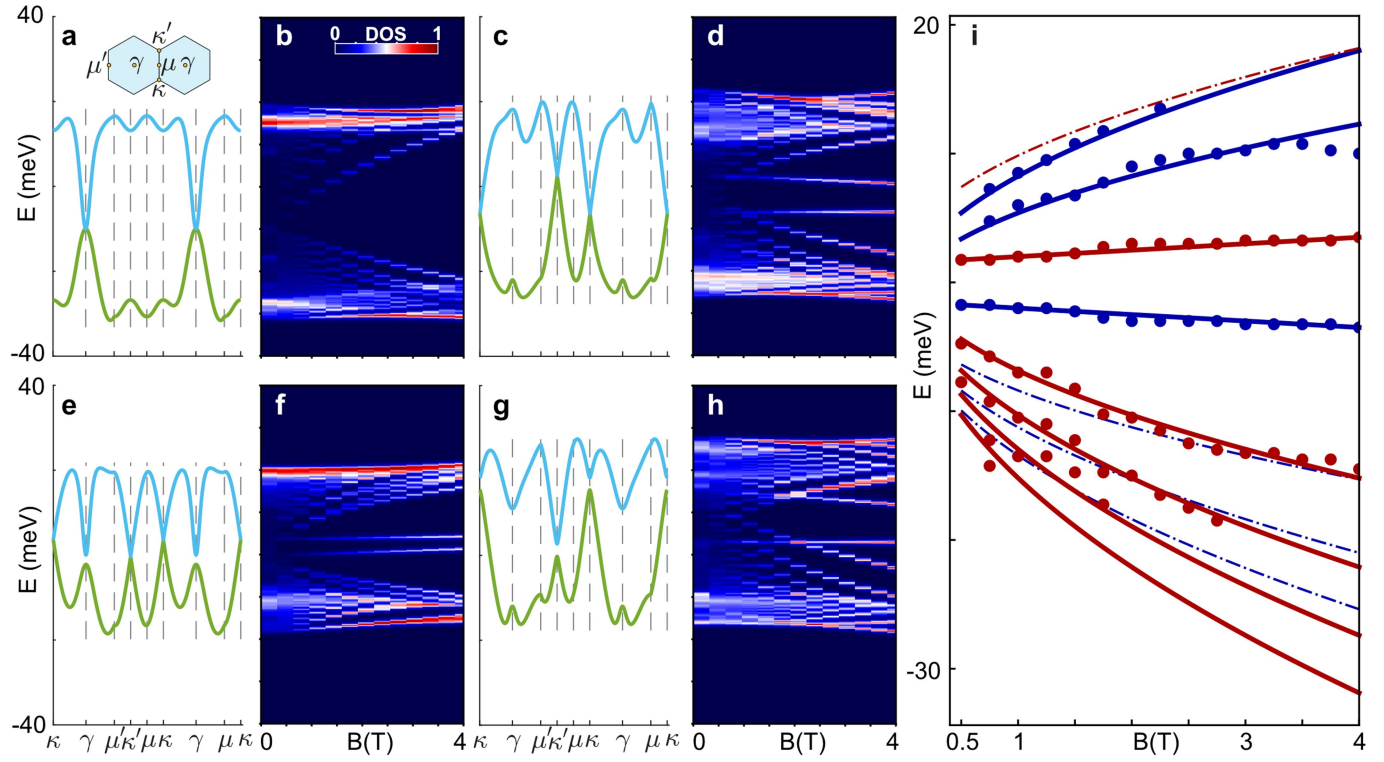
between VHSs) and exhibits an unrealistically strong dependence on the twist angle; even at angles slightly far from the magic angle ($\theta = 1.06$ – 1.09°), the model predicts a 2–4 times smaller bandwidth than that observed in the data. All spectra feature a well separated LL starting from each band edge that moves inwards with the field. This LL originates from the pocket near the centre of the moiré Brillouin zone and was not observed in the experiment. The bandwidth W_e (W_h) corresponding to the conductance (valence) flat VHS sub-band is used for estimating the U/W values in Extended Data Fig. 8.



Extended Data Fig. 8 | Estimated ratio between correlation and kinetic-energy scales, U/W , as a function of twist angle and magnetic field.

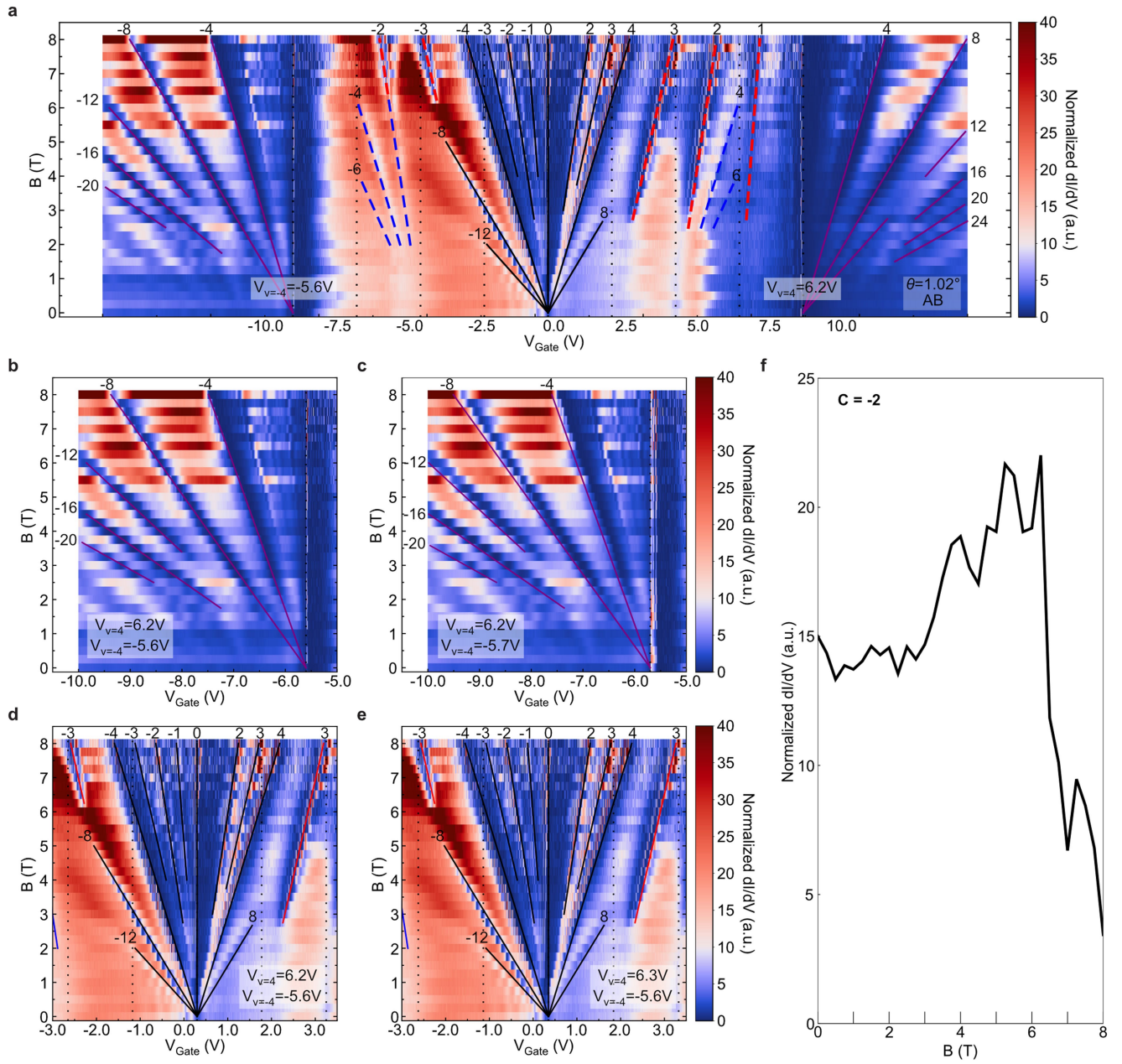
a, b, Estimates from the continuum model for the conduction (**a**) and valence (**b**) bands for $w_0/w_1 = 0.4$. **c, d,** The same calculation for $w_0/w_1 = 0.7$. **e,** Analytical estimate from equation (2). The bandwidth in **a, b, c, d** is evaluated as the total width of all levels that comprise the sub-band that includes the VHS (and carry $C = -1$ Chern number). Points very close to the magic angle are not shown owing

to a lack of clear energy separation from adjacent levels. The plotted constant- U/W curves outline regions where U/W exceeds a threshold value, and qualitatively reproduce the structure of the experimental phase diagram in which Chern insulating phases are observed (Fig. 3f). In **e**, the large η values are necessary to compensate for the crude approximation of the bandwidth from equation (3), which underestimates the bandwidth compared to the numerical simulation in **a, b, c, d**.



Extended Data Fig. 9 | Possible MATBG band-structure scenarios obtained using the ten-band model and corresponding LL spectra. a, b, Scenario (i); c, d, scenario (ii); e, f, scenario (iii); g, h, scenario (iv) (see Methods section ‘LL spectrum for low fields and MATBG band structure’). The inset in **a** shows the relevant high-symmetry momenta in the moiré Brillouin zone. Colours in **b, d, f** and **h** represent the normalized density of states (DOS) obtained via Lorentzian broadening of the Landau spectrum by 0.3 meV. **i**, A possible fit to the data in

Fig. 4j corresponding to scenario (ii), yielding $\nu_{D1} = 2.3 \times 10^5 \text{ m s}^{-1}$ and $\nu_{D2} = 1.9 \times 10^5 \text{ m s}^{-1}$ for the LLs originating from two Dirac cones (here marked with red and blue solid lines). The dashed lines correspond to expected LLs that were not observed, possibly owing to the relative positions of the Dirac cones with respect to the VHS. We note that this fitting scenario is not unique. However, all analysed scenarios give ν_D that exceeds 10^5 m s^{-1} .



Extended Data Fig. 10 | Landau fan diagram overlaid on experimental data for different choices of gate voltages corresponding to $\nu = \pm 4$. **a**, LDOS Landau fan with the same gate voltages $V_{V=4}$ and $V_{V=4}$, corresponding to $\nu = \pm 4$, as in Fig. 2c. The same gate voltages ($V_{V=4} = -5.6\text{V}$ and $V_{V=4} = 6.2\text{V}$) are used in **b** and **d**. **b, c**, LDOS Landau fan zoomed-in around the hole-side dispersive band with different choice of gate voltages. The choice in **c** ($V_{V=4} = -5.7\text{V}$ and $V_{V=4} = 6.2\text{V}$) does not reproduce the data well. **d, e**, Similar comparison around

the CNP, where **e** corresponds to $V_{V=4} = -5.7\text{V}$ and $V_{V=4} = 6.3\text{V}$. From this comparison, we extract the error in determining $\nu = \pm 4$ to be 0.1 V. All LDOS data are normalized by an average LDOS value for each magnetic field. **f**, Linecut of the LDOS fan in **a** along the LL filling of -2 emanating from $\nu = -2$, averaged over a V_{gate} window of 0.04 V. The state above $B = 6\text{ T}$ that we identify as a Chern insulating phase is distinguished from half-filling LLs by the abrupt drop in LDOS.